

Semantic Equivalence and Normal Forms

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Semantic Equivalence

Normal Forms

Definition

Definition

Let F be a set of propositional formulae, and φ a formula.

We say φ follows from F (or φ is a semantic consequence of F) if for all interpretations $\mathcal{I}$:

$$\mathcal{I}(\psi) = 1 \text{ for all } \psi \in F \implies \mathcal{I}(\varphi) = 1.$$

In symbols: $F \models \varphi$.

Example 1

$$\{p \rightarrow q, \neg q\} \models \neg p$$

Check using truth tables:

p	q	$p \rightarrow q$	$\neg q$	$\neg p$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	0
1	1	1	0	0

do the corresponding
columns contain 1s?

look for rows
containing only 1s

Example 2

Claim: $\{p \rightarrow q, \neg p\} \not\models \neg q$

($\neg q$ does *not* follow from $p \rightarrow q$ and $\neg p$)

First, let's have a look at the truth table:

p	q	$p \rightarrow q$	$\neg p$	$\neg q$
0	0	1	1	1
0	1	1	1	0
1	0	0	0	1
1	1	1	0	0

Let $\mathcal{I}$ be the interpretation with $\mathcal{I}(p) = 0$ and $\mathcal{I}(q) = 1$. Then, $\mathcal{I}(p \rightarrow q) = \mathcal{I}(\neg p) = 1$, but $\mathcal{I}(\neg q) = 0$. This proves the claim.

Reduction to satisfiability

Theorem

Let $\psi_1, \dots, \psi_n, \varphi$ be propositional formulae. Then:

$$\{\psi_1, \dots, \psi_n\} \models \varphi \iff \psi_1 \wedge \dots \wedge \psi_n \wedge \neg \varphi \text{ is unsatisfiable.}$$

Corollary:

For all propositional formulae φ :

$$\emptyset \models \varphi \iff \neg \varphi \text{ is unsatisfiable.}$$

Proof of the Theorem (1/2)

“ $\implies$ ” Let $\{\psi_1, \dots, \psi_n\} \models \varphi$. We have to show that the formula

$$\chi := \psi_1 \wedge \dots \wedge \psi_n \wedge \neg\varphi$$

is unsatisfiable, i.e., that χ is false under all interpretations. To this end, let $\mathcal{I}$ be an interpretation.

Case 1: $\mathcal{I}(\psi_i) = 1$ for all $i \in \{1, \dots, n\}$.

Since $\{\psi_1, \dots, \psi_n\} \models \varphi$, and $\mathcal{I}(\psi_i) = 1$ for all $i \in \{1, \dots, n\}$, we have $\mathcal{I}(\varphi) = 1$. This implies $\mathcal{I}(\neg\varphi) = 0$, hence $\mathcal{I}(\chi) = 0$.

Case 2: $\mathcal{I}(\psi_i) = 0$ for some $i \in \{1, \dots, n\}$.

In this case, we immediately obtain $\mathcal{I}(\chi) = 0$.

Proof of the Theorem (2/2)

“ $\Leftarrow$ ” Suppose

$$\chi := \psi_1 \wedge \cdots \wedge \psi_n \wedge \neg\varphi$$

is unsatisfiable. We have to show that $\{\psi_1, \dots, \psi_n\} \models \varphi$, i.e., that for all interpretations $\mathcal{I}$:

$$\mathcal{I}(\psi_i) = 1 \text{ for all } i \in \{1, \dots, n\} \implies \mathcal{I}(\varphi) = 1.$$

To this end, let $\mathcal{I}$ be an interpretation, and assume that $\mathcal{I}(\psi_i) = 1$ for all $i \in \{1, \dots, n\}$. Then, $\mathcal{I}(\psi_1 \wedge \cdots \wedge \psi_n) = 1$.

Now, since χ is unsatisfiable, we have $\mathcal{I}(\chi) = 0$. This implies that $\mathcal{I}(\psi_1 \wedge \cdots \wedge \psi_n) = 0$ or $\mathcal{I}(\varphi) = 1$. Since we have just ruled out the first case, we must have $\mathcal{I}(\varphi) = 1$.

Example 1 Revisited

Claim: $\{p \rightarrow q, \neg q\} \models \neg p$

Proof: It suffices to show that $\varphi = ((\neg p \vee q) \wedge \neg q) \wedge \neg \neg p$ is unsatisfiable.

We use the tableau algorithm:

- 1 Start with constraint $\mathcal{C}_0 = \{\varphi\}$
- 2 Applying to $\mathcal{C}_0$ the $\wedge$ -rule twice and then the $\neg\neg$ -rule yields:
 $\mathcal{C}_1 = \{\varphi, (\neg p \vee q) \wedge \neg q, \neg\neg p, \neg p \vee q, \neg q, p\}$
- 3 We can only apply $\vee$ -rules to $\mathcal{C}_1$. Each of these applications yields a constraint containing a clash.
- 4 Hence, φ is unsatisfiable. □

Exercise

Use truth tables or the tableau algorithm (preferably) to check which of the following are true/false:

① $\{p_1 \wedge p_2\} \models p_1 \vee p_2$

② $\{p_1\} \models p_2$

③ $\{p_1 \wedge \neg p_1\} \models p_2$

Tautologies

Definition

A propositional formula φ is a **tautology** if it is **true under all interpretations**.

Example: $p_1 \vee \neg p_1$ is a tautology:

p_1	$\neg p_1$	$p_1 \vee \neg p_1$
0	1	1
1	0	1

Observation: For all propositional formulae φ :

φ is a tautology $\iff \neg\varphi$ is unsatisfiable.

Semantic Consequence $\leftrightarrow$ Tautologies

Theorem

Let $\psi_1, \dots, \psi_n, \varphi$ be propositional formulae. Then:

$$\{\psi_1, \dots, \psi_n\} \models \varphi \iff (\psi_1 \wedge \dots \wedge \psi_n) \rightarrow \varphi \text{ is a tautology.}$$

Corollary:

For all propositional formulae φ :

$$\emptyset \models \varphi \iff \varphi \text{ is a tautology.}$$

Semantic Equivalence

Definition

Propositional formulae φ and ψ are said to be **equivalent** if for all interpretations $\mathcal{I}$:

$$\mathcal{I}(\varphi) = \mathcal{I}(\psi)$$

(φ and ψ have the same truth value under every interpretation).

In symbols: $\varphi \equiv \psi$.

Observation: *For all propositional formulae φ and ψ :*

$$\begin{aligned}\varphi \equiv \psi &\iff \{\varphi\} \models \psi \text{ and } \{\psi\} \models \varphi \\ &\iff \varphi \leftrightarrow \psi \text{ is a tautology.}\end{aligned}$$

Fundamental Laws for Equivalences

Associative laws:

$$\varphi \wedge (\psi \wedge \chi) \equiv (\varphi \wedge \psi) \wedge \chi$$

$$\varphi \vee (\psi \vee \chi) \equiv (\varphi \vee \psi) \vee \chi$$

Commutative laws:

$$\varphi \wedge \psi \equiv \psi \wedge \varphi$$

$$\varphi \vee \psi \equiv \psi \vee \varphi$$

Distributive laws:

$$\varphi \wedge (\psi \vee \chi) \equiv (\varphi \wedge \psi) \vee (\varphi \wedge \chi)$$

$$\varphi \vee (\psi \wedge \chi) \equiv (\varphi \vee \psi) \wedge (\varphi \vee \chi)$$

Complement law:

$$\neg \neg \varphi \equiv \varphi$$

De Morgan's laws:

$$\neg(\varphi \wedge \psi) \equiv \neg \varphi \vee \neg \psi$$

$$\neg(\varphi \vee \psi) \equiv \neg \varphi \wedge \neg \psi$$

HW

Exercise 1.4 page 82

1. 2) b, c, g, i
2. 3) a, c
3. 4) a, b
4. 7) a, c
5. 12) a, c, e
6. 16) c, f

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Semantic Equivalence

Normal Forms

Disjunctive Normal Form

Definition

- A **literal** is a proposition symbol or the negation of a proposition symbol.
- A **conjunction of literals** is a propositional formula of the form $\ell_1 \wedge \cdots \wedge \ell_m$, where each ℓ_i is a literal.
- A propositional formula is in **disjunctive normal form (DNF)** if it has the form $\varphi_1 \vee \cdots \vee \varphi_n$, where each φ_i is a conjunction of literals.

Examples

- p_1 , $\neg p_1$, p_5 are literals and also in DNF
- $p_1 \wedge \neg p_2$ is in DNF
- $p_2 \vee (\neg p_2 \wedge p_3)$ and $(p_1 \wedge \neg p_4 \wedge p_6) \vee (\neg p_3 \wedge p_2)$ are in DNF
- $p_1 \wedge (\neg p_2 \vee p_3)$ and $p_1 \vee \neg(\neg p_2 \wedge p_3)$ are *not* in DNF

Conjunctive Normal Form

Definition

- A **disjunction of literals** is a propositional formula of the form $\ell_1 \vee \cdots \vee \ell_m$, where each ℓ_i is a literal.
- A propositional formula is in **conjunctive normal form (CNF)** if it has the form $\varphi_1 \wedge \cdots \wedge \varphi_n$, where each φ_i is a disjunction of literals.

Examples (cont'd)

- $p_1, \neg p_1, p_5$ are in DNF and CNF
- $p_1 \wedge \neg p_2$ is in DNF and CNF
- $\neg p_2 \vee (\neg p_2 \wedge p_3 \wedge p_7) \vee (p_3 \wedge \neg p_1)$ is in DNF, but *not in CNF*
- $\neg p_2 \wedge (\neg p_2 \vee p_3 \vee p_7) \wedge (p_3 \vee \neg p_1)$ is *not in DNF*, but in CNF

Exercise

Which of the following formulae is in DNF, CNF, or both, or none?

① $(p_1 \wedge \neg p_4 \wedge p_1) \vee (p_2 \wedge \neg p_4)$

② $p_1 \vee ((p_2 \vee p_3) \wedge p_4)$

③ $p_1 \vee \neg p_2$

Transformation into DNF and CNF

Theorem

Let φ be a propositional formula.

- 1 φ is equivalent to a formula in DNF.
- 2 φ is equivalent to a formula in CNF.

Note:

While checking satisfiability for formulae in DNF is easy, an equivalent formula in DNF may be exponentially longer than the original formula.

Transformation into DNF

Transformation of a formula φ into DNF:

- **Step 1:** Apply De Morgan's and the complement law to φ until negation occurs in front of proposition symbols only.

Let φ' be the resulting formula.

- **Step 2:** Apply distributive and commutative laws to φ' until the formula is in DNF.

Example

Goal: Transform $\varphi = \neg(p \wedge \neg q) \wedge (\neg r \vee p)$ into DNF

Step 1:

$$\begin{aligned}\varphi &\equiv (\neg p \vee \neg\neg q) \wedge (\neg r \vee p) && \text{(by De Morgan's law)} \\ &\equiv \underbrace{(\neg p \vee q) \wedge (\neg r \vee p)}_{\varphi'} && \text{(by complement law)}\end{aligned}$$

Step 2:

$$\begin{aligned}\varphi' &\equiv ((\neg p \vee q) \wedge \neg r) \vee ((\neg p \vee q) \wedge p) && \text{(by distributive law)} \\ &\equiv (\neg r \wedge (\neg p \vee q)) \vee (p \wedge (\neg p \vee q)) && \text{(by commutative law)} \\ &\equiv ((\neg r \wedge \neg p) \vee (\neg r \wedge q)) \vee ((p \wedge \neg p) \vee (p \wedge q)) && \text{(by distributive law)} \\ &\equiv \underline{(\neg r \wedge \neg p) \vee (\neg r \wedge q) \vee (p \wedge \neg p) \vee (p \wedge q)} && \text{(in DNF)}\end{aligned}$$

Transformation into CNF

The DNF transformation method works (without change) for CNF!

Transformation of a formula φ into CNF:

- **Step 1:** Apply De Morgan's and the complement law to φ until negation occurs in front of proposition symbols only.

Let φ' be the resulting formula.

- **Step 2:** Apply distributive and commutative laws to φ' until the formula is in CNF.

Example

Goal: Transform $\varphi = \neg(p \vee \neg q) \vee (\neg r \wedge p)$ into CNF

Step 1:

$$\begin{aligned}\varphi &\equiv (\neg p \wedge \neg\neg q) \vee (\neg r \wedge p) && \text{(by De Morgan's law)} \\ &\equiv \underbrace{(\neg p \wedge q) \vee (\neg r \wedge p)}_{\varphi'} && \text{(by complement law)}\end{aligned}$$

Step 2:

$$\begin{aligned}\varphi' &\equiv ((\neg p \wedge q) \vee \neg r) \wedge ((\neg p \wedge q) \vee p) && \text{(by distributive law)} \\ &\equiv (\neg r \vee (\neg p \wedge q)) \wedge (p \vee (\neg p \wedge q)) && \text{(by commutative law)} \\ &\equiv ((\neg r \vee \neg p) \wedge (\neg r \vee q)) \wedge ((p \vee \neg p) \wedge (p \vee q)) && \text{(by distributive law)} \\ &\equiv (\neg r \vee \neg p) \wedge (\neg r \vee q) \wedge (p \vee \neg p) \wedge (p \vee q) && \text{(in CNF)}\end{aligned}$$

HW

Exercise 1.5 page 87

1. 2) a, d
2. 7) a, b
3. Extra exercises: Convert the following into *CNF*:
 - 3.1 $(p \rightarrow q) \leftrightarrow (p \rightarrow r)$
 - 3.2 $(p \vee q) \wedge (p \wedge r)$
 - 3.3 $(p \rightarrow q) \rightarrow r$
 - 3.4 $p \leftrightarrow q$
 - 3.5 $(a \rightarrow b) \rightarrow ((b \rightarrow c) \rightarrow (a \rightarrow c))$

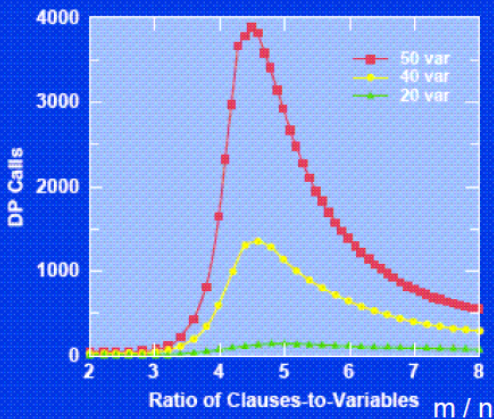
An overview of the satisfiability problem (SAT)

Fixed-length clause model

- ▶ *k-SAT*: each clause contains exactly k literals
- ▶ Suppose we have m clauses in n variables
- ▶ Randomly generate the problem for fixed k and varying m and n .

Question: How does the problem hardness vary with $\frac{m}{n}$?

3-SAT Hardness



As n increases
hardness
transition
grows sharper

Threshold conjecture

- ▶ For every k , $\exists c^*$ such that
 - ▶ $\frac{m}{n} < c^*$, as $n \rightarrow \infty$ problem is *satisfiable* with prob 1.
 - ▶ $\frac{m}{n} > c^*$, as $n \rightarrow \infty$ problem is *unsatisfiable* with prob 1.
- ▶ The conjecture is true for $k = 2$, namely $c^* = 1$.
- ▶ For $k = 3$, currently know $3.42 \leq c^* \leq 4.51$

$(2 + p)$ -SAT model

- ▶ 2SAT is in P
- ▶ 3SAT is in NP
- ▶ What if a problem is a mixture of 2 & 3 - clauses. Suppose
 - ▶ $1 - p$: fraction of 2 - clauses.
 - ▶ p : fraction of 3 - clauses.
- ▶ This is the so-called $2 + p$ - SAT problem
- ▶ Experimentation with random $(2 + p)$ -SAT:
 - ▶ When $p < 0.41$, the problem behaves like 2-SAT
 - ▶ When $p > 0.42$, the problem behaves like 3-SAT